PROJECT 10: DELIVERY SERVICE MANAGEMENT SYSTEM

DATCOM Lab
NEU-College of Technology
National Economics University
Email: hung.tran@neu.edu.vn

Project Objective

The objective of this project is to develop a database-driven system that supports the management of delivery services, including customer orders, delivery assignments, vehicle tracking, and expense management, in order to enhance operational efficiency and transparency.

Database and Programming Languages

- **Database Management System:** MySQL
- **Programming Language:** Python

Detailed Project Requirements

System Analysis and Requirements

- Manage detailed information about customers, delivery orders, transport vehicles, and delivery-related expenses.

Main Functionalities

- Customer management (add/update customer profiles and contact information).
- Order management (create, track, update delivery orders).
- Delivery scheduling and tracking (assign vehicles and update status).
- Vehicle management (record type, license plate, and availability).
- Expense tracking for each delivery (fuel, tolls, handling fees, etc.).
- Generate reports on delivery performance, cost summaries, and order status.

Database Design and Implementation

1. **Data Model Design**
   - Design an ER diagram to reflect relationships among orders, vehicles, customers, and deliveries.
   - Convert ER diagram into relational schema with PKs, FKs, and constraints.
2. **Table Structures**
   - Customers (CustomerID, CustomerName, PhoneNumber, Address)
   - Orders (OrderID, CustomerID, OrderDate, Status)
   - Deliveries (DeliveryID, OrderID, VehicleID, DeliveryDate)
   - Vehicles (VehicleID, VehicleType, LicensePlate)
   - Expenses (ExpenseID, DeliveryID, ExpenseType, Amount)
3. **Sample Data**
   - Insert 510 rows of sample data for each table.
   - Use MySQL Workbench to generate the database schema diagram.

Advanced Database Objects

- **Indexes:** Speed up queries for order lookup, delivery status, and vehicle availability.
- **Views:** Create views for current delivery schedules, cost per order, or outstanding orders.
- **Stored Procedures:** Automate delivery assignment or calculate total expenses.
- **User Defined Functions:** Compute average delivery cost or number of deliveries per vehicle.
- **Triggers:** Automatically update order status when delivery is confirmed.

Database Security and Administration

- Define roles for delivery managers, dispatchers, and accountants.
- Apply access control and data encryption for sensitive information.
- Set up backup and recovery procedures.
- Use query optimization to improve real-time tracking and reporting.

Python Application Development

- **Database Connection:** Use `mysql-connector-python` or `SQLAlchemy`.
- **Order and Delivery Management:** Build functions to manage orders, assign deliveries, and generate invoices.
- **Reports:** Generate performance reports, delivery history, and cost breakdowns.
- **User Interface:** Create a console-based or GUI app to interact with the delivery system.

Deliverables

- A printed comprehensive report including database design, implementation details, and screenshots. Submit your report into LMS. First pages of your report must attach this PDF file.
- Link to publish Github source code and video presenting on Youtube
- SQL schema, sample data scripts, ER diagrams, and Python code.
- Demonstration of functionalities such as adding income, managing expenses, and generating reports.

Conclusion and Recommendations

- Summarize key outcomes and effectiveness of the delivery management system.
- Recommend additional features such as GPS integration, SMS/email notifications, or delivery rating.

References

- List all resources, datasets, and documentation referenced during the project.